

DUSTRUNNER · DESIGN PRECIS

DustRunner design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	DRN-PRC-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/dustrunner		
Source	docs/02-concept.md @ f77845b		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, traction, wind, economics, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

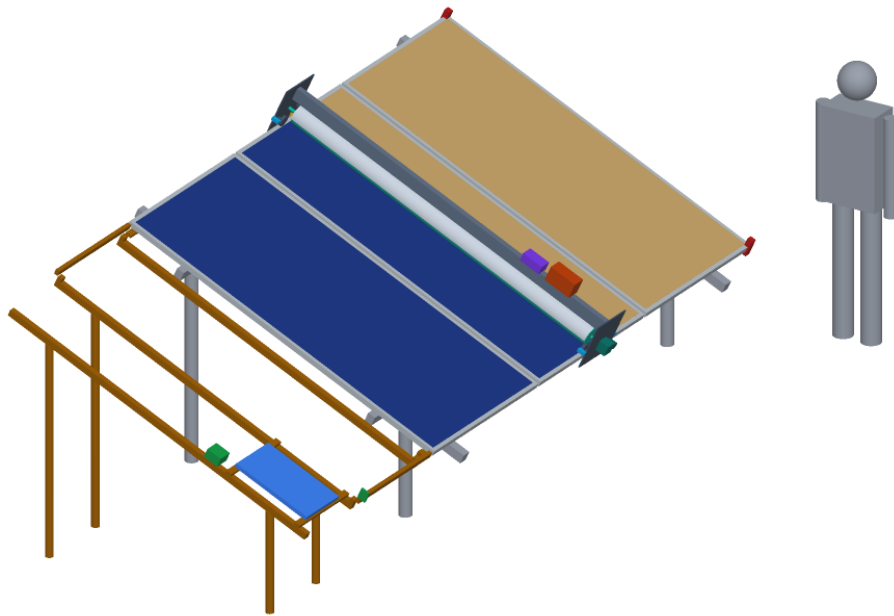
DustRunner design precis

■ Summary

DustRunner is a rail-free crawler robot that spans one PV table from its lower to its upper module edge, rides along the module frames on two end trucks that hook under the frame lips, and sweeps the glass dry with a 2.2 m rotating microfiber brush. It parks off the glass in a clamp-on dock at the start of the row, where a 20 W panel recharges its 12.8 V LiFePO4 pack. First-order estimates for a 40 m, 20 kWp reference row: about 7 minutes and 8 Wh per out-and-back cycle, a robot of about 12 kg, and about \$485 in parts for robot and dock. The weak points are traction on dusty frames, wind, unproven cleaning effectiveness and glass abrasion, and a payback of about 4 years in the mid case (R14 not met).

DustRunner

CONCEPT, NOT FOR FABRICATION



Grey: 1.75 m person for scale

Figure 1. DustRunner partway along a three-module section of the reference table, moving away from its dock (brown, left) toward the end stops (red). Glass behind the robot is clean; tan shows dust still to be removed. Grey figure: 1.75 m person for scale.

■ How it works

1. **Park and charge.** Between cycles the robot sits in the dock, a short pair of rails that continue the module edges past the end of the row. Parking off the glass means the robot never shades the array. A 20 W panel in the dock charges the pack through spring contacts.
2. **Start.** Once a day, after the array's output has dropped in the evening and before dew forms (proposed, awaiting Amish), the controller checks pack charge, air humidity and a wind limit, then drives out of the dock.
3. **Travel.** Two end trucks carry the robot along the row. Each truck has two polyurethane wheels on the top of the module frame, two guide rollers against the outer face of the frame and a spring-loaded hook roller under the frame lip. The trucks are the "clamp": they cannot lift off or slide down the slope, yet need no rail. A drive gearmotor with an encoder on each truck drives both of that truck's wheels through a toothed belt, and the controller keeps the two ends in step.
4. **Clean.** A 120 mm microfiber brush on a 2.2 m aluminum tube spans the slope and turns at about 150 rpm, lifting dust ahead of the robot. A helical pattern on the brush conveys the loosened dust down the slope and off the lower edge. A hood over the brush limits dust thrown back onto clean glass.
5. **Stop and return.** IR sensors on each truck see the frame end; a clamp-on end stop at the far end is the mechanical backstop. The robot reverses the brush and cleans on the way back, then docks.

DustRunner: energy per cleaning cycle, 40 m row out and back, Wh (estimates)

CONCEPT, NOT FOR FABRICATION

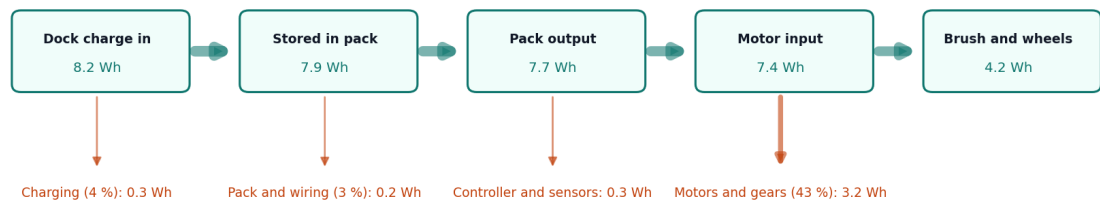


Figure 2. Energy for one out-and-back cycle on the 40 m reference row, in Wh. All values are estimates.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv` .

#	Component	Proposed choice	Notes
1	Chassis beam	40 x 80 mm aluminum rectangular tube, about 2.3 m	Cut to the module length for other tables
2	Microfiber brush	120 mm diameter, 2.2 m, on a 40 mm aluminum tube	Replaceable microfiber sleeves with a helical pattern
3	Brush hood	Bent 1 mm aluminum or polycarbonate half-shell	Also the finger guard at the brush ends
4	Brush motor and belt drive	12 V DC gearmotor, about 60 W, toothed belt	Reversible; about 150 rpm at the brush
5	End trucks and wheels	6 mm aluminum plates, two 70 mm polyurethane wheels each, belt-linked	Wheels run on the frame top only
6	Drive gearmotors	Two 12 V gearmotors with encoders, one per truck	Synchronized in firmware
7	Guide and hook rollers	Sealed-bearing rollers: guide rollers on the frame face, spring-loaded hook rollers under the lip	Keep the robot on the table in wind
8	Battery	LiFePO ₄ , 12.8 V, 10 Ah (128 Wh) with BMS	Chemistry chosen for heat tolerance and safety
9	Controller	ESP32-class board, two drive channels, brush driver, current sensing, humidity and temperature sensor, IP65 box	Firmware is a sketch until TRL 4
10	IR edge sensors	Four downward-looking reflective sensors, two per truck	Row-end and gap detection
11	Dock frame	Aluminum angle rails continuing the module edges, cross members, two clamps to the first module frame and four legs	Clamp-on; no drilling
12	Dock PV panel	20 W, about 530 x 350 mm, in the table plane	Charges the parked robot
13	Dock charger and contacts	LiFePO ₄ charge controller and spring contacts	
14	End stops	Pair of clamp-on blocks on the last module's frame	Mechanical backstop

DustRunner: exploded view (robot laid flat; dock and end stop drawn beside it)

CONCEPT, NOT FOR FABRICATION

- 1 Chassis beam, 40 x 80 mm
- 2 Microfiber brush, 120 mm x 2.2 m
- 3 Brush hood
- 4 Brush motor and belt drive
- 5 End trucks and wheels (pair)
- 6 Drive gearmotors (2)
- 7 Guide and hook rollers (set)
- 8 LiFePO4 pack, 12.8 V 10 Ah
- 9 Controller and motor drivers
- 10 IR edge sensors (4)
- 11 Dock frame, rails and clamps
- 12 Dock PV panel, 20 W
- 13 Dock charger and contacts
- 14 End stops (pair)

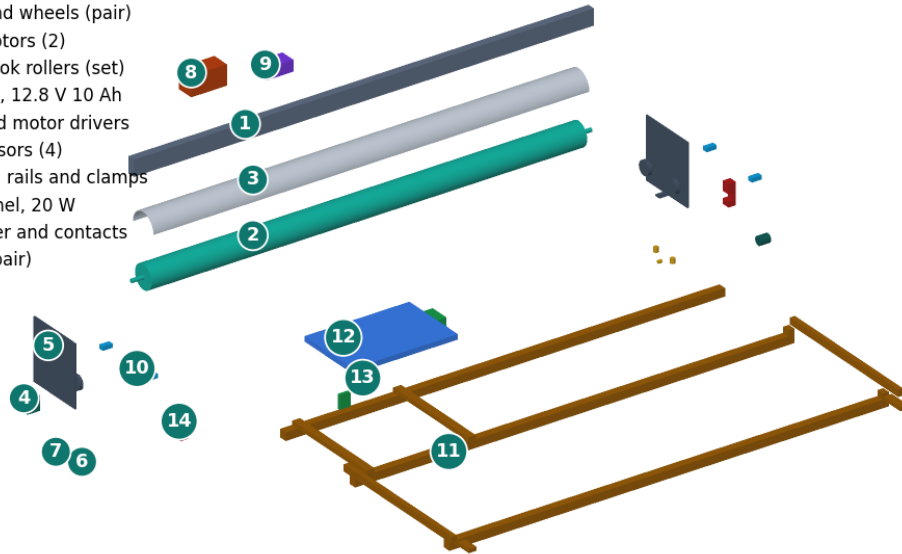
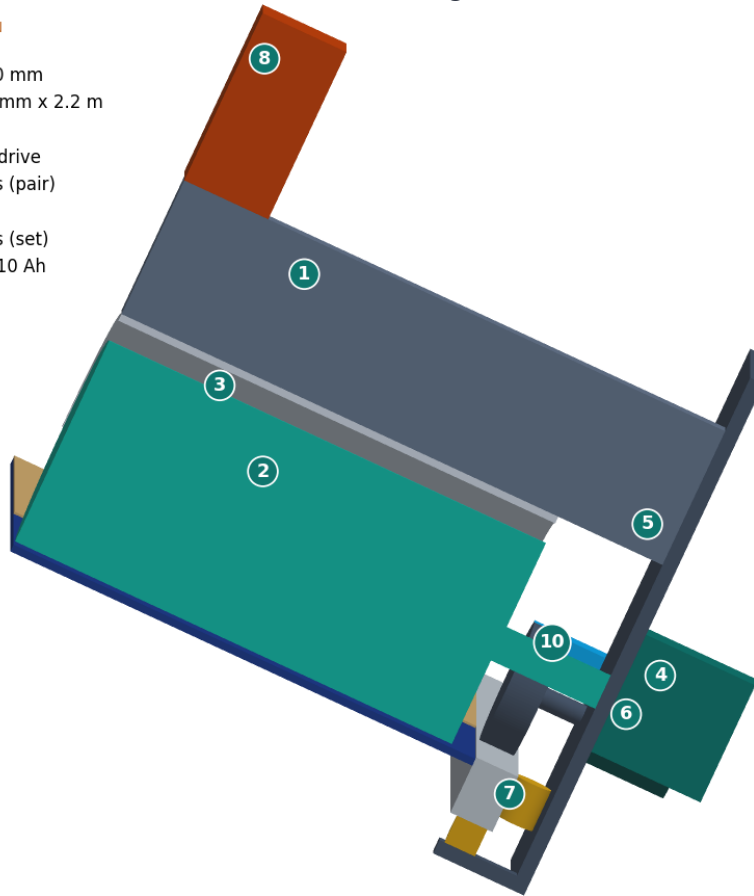


Figure 3. Exploded view with numbered callouts matching the BOM. The robot is laid flat in the table plane; the dock and end stops are drawn beside it. Upper-end twins of paired parts are shown without numbers.

DustRunner: section across the row at the lower module edge

CONCEPT, NOT FOR FABRICATION

- 1 Chassis beam, 40 x 80 mm
- 2 Microfiber brush, 120 mm x 2.2 m
- 3 Brush hood
- 4 Brush motor and belt drive
- 5 End trucks and wheels (pair)
- 6 Drive gearmotors (2)
- 7 Guide and hook rollers (set)
- 8 LiFePO4 pack, 12.8 V 10 Ah
- 10 IR edge sensors (4)



Silver: module frame; blue: glass. Truck (5) wheels ride the frame top; rollers (7) bear on the frame face and hook under its lip

Figure 4. Section across the row through the robot at the lower module edge, showing how the end truck wraps the frame.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Reference row and robot mass

Quantity	Estimate	Basis
Reference row	About 40 m, 35 modules, about 90 m ² of glass, about 20 kWp	35 x 1,154 mm pitch; 2,278 x 1,134 mm modules at about 580 W
Robot mass	About 12 kg	Beam 3.1 kg, brush 2.3, trucks and wheels 2.0, motors 1.2, battery 1.3, hood 0.7, rollers 0.4, controller 0.4, sensors and wiring 0.5
Load normal to the glass	About 107 N	12 kg x 9.81 m/s ² x cos 25°
Load down the slope	About 50 N	12 kg x 9.81 x sin 25°, carried by the upper truck's guide rollers
Wheel load	About 27 N, or about 52 N with 50 N of hook preload per truck	(107 N + 100 N) / 4 wheels

The wheel loads are small next to the loads modules are rated for (for example the 2,400 Pa test load on about 2.6 m² is about 6 kN), but they act on the frame edge, so the module maker's guidance on frame loading must be checked (R10).

Power and energy per cycle

Assumptions: travel 0.2 m/s; brush surface speed about 0.94 m/s (120 mm at 150 rpm); brush drag about 30 W of mechanical power (the largest uncertainty, to be measured); gearmotor and belt efficiency 60 % for the brush and 50 % for the drives; controller and sensors 3 W.

Quantity	Estimate	Basis
Brush tangential force	About 32 N	30 W / 0.94 m/s
Drive force, worst case	About 39 N	Rolling 2 N (0.02 x 107 N), brush reaction 32 N, 2° row grade 4 N, roller drag 1 N
Power at the wheels	About 8 W	39 N x 0.2 m/s
Electrical power while cleaning	About 70 W	Brush 50 W, drives 16 W, electronics 3 W
Cycle time, 40 m row out and back	About 7 min	80 m / 0.2 m/s = 400 s
Pack energy per cycle, 40 m row	About 7.7 Wh (8.2 Wh from the dock)	69 W x 400 s
Cycle time and energy, 100 m row	About 17 min, about 20 Wh from the dock	R7 met
Cleaning rate	About 1,600 m ² per hour of travel	2.2 m x 0.2 m/s

Charging and autonomy

Quantity	Estimate	Basis	Requirement
Usable pack energy	About 102 Wh	128 Wh x 80 %	
Days of cleaning with no sun, 40 m row	About 13	102 / 7.7 Wh	
Days of cleaning with no sun, 100 m row	About 5	102 / 19.2 Wh	R8 met
Dock panel harvest	About 82 Wh/day at 5.5 sun hours; about 45 Wh/day at 3	20 W x hours x 0.75	R8 met

The pack is cycled lightly (under 10 % of capacity a day on the reference row), which is kind to LiFePO4 cells in heat. A smaller 6 Ah pack would work on rows under about 60 m; the 10 Ah pack is kept for longer rows and cloudy spells.

Traction

Traction is the tightest mechanical margin. If only one wheel per truck were driven, the driven wheels would carry about 54 N, and at an assumed friction coefficient of 0.4 for polyurethane on a dusty anodized frame they could push only about 21 N against the 39 N needed. Driving both wheels on each truck and preloading the hook rollers against the frame lip with springs raises the normal force to about 207 N and the available traction to about 83 N, a margin of about 2. The friction coefficient on dusty frames is an assumption and must be measured.

Wind

Case	Estimate	Consequence
Along-row wind at 8 m/s while cleaning	About 34 N on about 0.58 m ² of robot side area (drag coefficient 1.5)	Total drive force about 73 N, within the 83 N available; stop cleaning above 8 m/s
Parked in the dock at 35 m/s	About 650 N along the row; about 280 N of lift against about 107 N of weight	Dock latch and hook rollers must hold about 650 N along and about 170 N up

A wind limit therefore needs a wind input: an anemometer at the dock (about \$15 more), or a weather feed. This is an open question.

Energy recovered, water saved and payback

Assumptions are listed in DRN-REQ-001: 5.5 kWh per kWp per day, a manual wash every 30 days as the baseline, \$0.10/kWh, and an average residual loss of about 1.5 % with daily dry cleaning (unverified, R2).

Case	Average loss, monthly wash	Average loss, daily DustRunner	Recovered energy	Value	Simple payback on \$485
Mid, 0.3 % per day, 40 m row	About 4.5 %	About 1.5 %	About 1,220 kWh a year	About \$120 a year	About 4.0 years (R14 not met)
High, 0.49 % per day, 40 m row	About 7.4 %	About 1.5 %	About 2,390 kWh a year	About \$240 a year	About 2.0 years
Mid, 100 m row	About 4.5 %	About 1.5 %	About 3,030 kWh a year	About \$300 a year	About 1.6 years

Water saved is about 120 to 350 L per avoided wash of the reference row (3.5 to 10 L per module), about 1.5 to 4 m³ a year for monthly washing. The case is strongest for long rows, for sites that currently do not clean at all, and for mini-grids where each kilowatt-hour displaces diesel.

Cost

Group	Indicative cost
Robot (items 1 to 10)	About \$353
Dock and end stops (items 11 to 14)	About \$102
Wiring, fuse, stop buttons and hardware (item 15)	About \$30
Total	About \$485 (R13 met, about 3 % margin)

■ Key design choices

Every choice below is **Proposed, awaiting Amish**.

- **One robot per row, spanning the full slope.** A full-width robot needs no steering on the glass and cannot fall between modules. The alternatives are a smaller robot that zigzags on the glass (needs edge sensing on all sides and loads the glass) or a transfer cart that carries one robot between rows (more cost and complexity). Recommendation: one full-width robot per row.
- **Ride on the frame edges, not on the glass.** Wheels on the frame keep weight off the glass and coating. This limits the concept to tables whose short edges are free of clamps (1P portrait with clamps on the long sides, as on the reference table). Recommendation: frame riding, with the fit limit stated in R4.
- **Rail-free, with a short dock and clamp-on end stops.** The dock and stops are the only added hardware and clamp on without drilling. Fully rail-free operation with no end stop would rely on the IR sensors alone. Recommendation: keep the mechanical end stops as the independent backstop (R9).
- **Dry microfiber brush, no water and no airflow.** Microfiber is the gentlest common dry medium; an airflow stage (as Ecoppia uses) would add a fan and power. Recommendation: microfiber brush first; airflow only if tests show dust is re-deposited.
- **Charging at the dock, not from a panel on the robot.** A panel on the robot (as Ecoppia uses) needs no contacts but adds mass and wind area. Recommendation: 20 W dock panel.
- **Both wheels driven on each truck, with preloaded hook rollers.** See Traction. Recommendation: belt-linked wheels and spring preload.

- **Cleaning time.** Options: at night (no shading, cool glass, but dew can turn dust to mud), in the evening before dew forms (low shading loss), or at dawn (worst dew). While on the glass the robot shades a strip along one module, which can bypass a cell substring for a minute or two. Recommendation: evening, gated by the humidity sensor.
- **LiFePO4 at 12.8 V.** Safer and more heat-tolerant than lithium-ion NMC, and 12 V motors and chargers are common. Recommendation: 12.8 V LiFePO4. A SwapCell pack is not proposed: at about 468 Wh and 48 V it is far larger than DustRunner needs.
- **Budget.** Indicative parts cost is about \$485 against the \$500 budget; an anemometer would bring it to about \$500. `project.yaml` is unchanged.

■ Safety

Safety: DustRunner is moving machinery with a rotating brush, a lithium iron phosphate battery, and it works on an energized PV array that can carry several hundred volts DC, on hot glass, up to about 1.6 m above the ground.

- **Rotating brush and pinch points.** The brush, belts and wheels can catch fingers, hair and clothing. The hood covers the brush ends; each truck has an emergency stop button; the brush stops when a truck lifts off the frame. Nobody should reach under the hood while the robot is powered.
- **Unexpected start.** The robot runs on a timer with nobody present. It must carry a clear warning label, start only after a short audible warning, and have a lockout switch at the dock.
- **Falling robot.** A 12 kg robot leaving the row end could injure someone below. Two independent stops (IR sensing and the mechanical end stop) and the hook rollers address this (R9); the design must be checked for wind at TRL 3.
- **PV array electrical hazard.** Strings carry DC voltages that can be lethal and cannot be switched off in daylight. Nothing on the robot may touch cables, connectors or junction boxes; installation and maintenance follow the array owner's electrical safety rules. Damaged module glass exposes live parts.
- **Battery.** LiFePO4 is less prone to thermal runaway than other lithium-ion chemistries but can still vent and burn if crushed, shorted or overcharged. Use a pack with a BMS, fuse the pack output, block charging below 0 °C, and keep the pack shaded and ventilated.
- **Hot surfaces.** Module glass and frames can reach about 75 °C. Wear gloves when installing or servicing in daylight.
- **Module warranty.** Abrasive cleaning, or a cleaning tool the maker has not cleared, can void the module warranty. Check the maker's cleaning guidance and ask the maker to clear the robot before use (see DRN-PRB-001).

■ Open questions

- [] Brush drag power and cleaning effectiveness on real dust, dry and after dew (R2).
- [] Long-term abrasion of anti-reflective coatings by daily microfiber contact (R3).
- [] Friction coefficient of polyurethane on dusty frames; spring preload value (Traction).
- [] Wind input for the operating limit: anemometer at the dock or weather feed?
- [] Module makers' positions on frame loading and robotic dry cleaning.
- [] How dust conveyed off the lower edge builds up at the lower frame lip over time.

- [] Pilot site, partner and first table format (see DRN-PRB-001).